

Sarthak Singh

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EDUCATION

Stony Brook University

Master of Science in Computer Science — Teaching Assistant: Object Oriented Programming

Stony Brook, NY

Aug. 2024 to May 2026

NMIMS MPSTME

Bachelor of Technology in Computer Engineering

Mumbai, India

Aug. 2020 to May 2024

EXPERIENCE

AI Systems Engineer

Jul. 2026 to Present

Brave Federal Technology LLC

McLean, VA — customer.bravefederal.com

- Spearheaded BraceCX, one of 2 BCX deliverables, taking the product from initial implementation through production rollout, operational handoff, and repeatable release workflows.
- Engineered a migration pipeline across 6 integrations from Avaya, Nortel, and Cisco into Webex, Teams, and Slack, supporting repeatable onboarding and system transition workflows.
- Implemented validation and transformation checks that standardized onboarding data, improved import consistency, and reduced manual remediation when migration failures occurred.

Graduate Research Assistant

Jan. 2025 to Dec. 2025

Stony Brook University

Stony Brook, NY

- Architected a GPU accelerated 3D reconstruction engine in C++ and Python with CUDA, reaching 25 to 30 dB PSNR for repeated scene rendering and model evaluation workloads.
- Redesigned a modular pipeline for 5 input formats, improving throughput by 40% through memory efficient access patterns, streamlined data flow, and reusable processing stages.

Software Engineering Intern

Dec. 2023 to May 2024

NMIMS University

Mumbai, India

- Delivered a Java and Python backend handling 50K+ daily requests at 89% accuracy for university document workflows, internal processing, and service integrations.
- Scaled request handling to 10,000+ concurrent connections with asynchronous I/O and thread pooling, cutting tail latency by 25% under load tests.
- Instrumented monitoring across 500+ sessions to isolate query and endpoint bottlenecks, driving two rounds of targeted performance optimization.

Software Engineering Intern

May 2023 to Aug. 2023

Vodafone Idea Ltd.

Mumbai, India

- Engineered auto scaling cloud infrastructure with resource monitoring in Java, reducing CPU overhead by 15% during peak usage and release deployment windows.
- Orchestrated zero downtime migration of 12+ microservices, cutting outages by 20% through phased rollout, validation, rollback planning, and service checks.
- Introduced CI/CD regression tests that caught degradation before deployment and raised production reliability by 25% across application releases.

PROJECTS

Enterprise RAG System with Multi Document QA | Python, FastAPI, FAISS, Pinecone, Redis, OpenAI | [GitHub](#) Dec. 2025

- Architected a production RAG pipeline processing 10K+ documents with FAISS and Pinecone indexing, reaching 92% answer accuracy and 0.87 precision at 5.
- Engineered a FastAPI REST service with hybrid search, Redis caching, citation tracking, and Prometheus metrics, handling 1,200+ queries per hour at 240ms latency.

Distributed Banking System with Consensus Protocol | Python, gRPC, Docker, Redis

Sep. 2023 to Dec. 2023

- Designed a fault tolerant transaction engine using Linear PBFT consensus across 10+ nodes, sustaining 99.9% uptime and 1,000+ TPS with Byzantine resilience.
- Optimized RPC communication to reduce tail latency from 500ms to 320ms, a 36% gain, through multiplexed channels and compact state serialization.

Ring All Reduce Benchmark | Python, PyTorch, torch.distributed, Gloo, NCCL | [GitHub](#)

Aug. 2026

- Implemented ring all reduce from scratch with torch.distributed point to point communication, using scatter reduce and all gather phases across 4 worker processes.
- Benchmarked manual collective communication against PyTorch dist.all.reduce across 1 KB to 256 MB tensors, collecting bandwidth results and exposing latency overhead at smaller message sizes.

TECHNICAL SKILLS

Languages: C++, Python, Java, Go, SQL, Bash

Backend and Systems: Linux, TCP/IP, Socket Programming, REST APIs, gRPC, Microservices, Distributed Systems, System Design, Multithreading, Concurrency, Fault Tolerance, Debugging

Platform and Infrastructure: Docker, Kubernetes, AWS, PostgreSQL, Redis, MongoDB, Terraform, Jenkins, CI/CD, CMake, CTest

Data and AI: FastAPI, FAISS, Pinecone, OpenAI API, RAG, LangChain